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Vehicle steering assembly

Abstract

A vehicle steering assembly for controlling movement of a vehicle having independently rotatable left and right ground-engaging traction elements. The steering assembly comprises a steering handle and a steering controller positioned generally below the steering handle and actuated by movement of the steering handle. The steering handle includes an upright shaft and a laterally-extending crossmember fixed to the top of the upright shaft. A flexible protective cover is attached to the shaft and is configured to prevent dirt and debris from entering the steering controller. The steering handle is shiftable in forward and rearward directions to thereby cause corresponding forward and rearward rotation of both of the left and right traction elements. The steering handle is rotatable in clockwise and counterclockwise directions to thereby cause a change in the relative speeds and directions of rotation of the left and right traction elements.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present non-provisional patent application is a continuation-in-part patent application, which claims priority to U.S. patent application Ser. No. 17/721,544, filed on Apr. 15, 2022, and entitled “VEHICLE STEERING ASSEMBLY,” which claims priority to U.S. Pat. No. 11,305,806, filed on Aug. 14, 2019, and entitled “VEHICLE STEERING ASSEMBLY,” which claims priority to U.S. Provisional Patent Application Ser. No. 62/718,801, filed on Aug. 14, 2018, and entitled “VEHICLE STEERING ASSEMBLY,” with the entirety of the above-identified, previously-filed provisional application being hereby incorporated by reference into the present non-provisional patent application.

FIELD OF THE INVENTION

(1) Embodiments of the present invention are generally directed to a vehicle steering assembly. More particularly, embodiments of the present invention are directed to an improved vehicle steering assembly for a compact utility loader.

BACKGROUND OF THE INVENTION

(2) Compact utility loaders are becoming popular options for operators in need of heavy equipment machines. Compact utility loaders are capable of operating a variety of different hydraulically-driven tools or attachments for performing various types of demanding work. Beneficially, however, compact utility loaders are generally manufactured with a relatively smaller size compared to other heavy equipment machines, which can be beneficial for maneuverability, transport, and storage. Unfortunately, the control systems of previously-used compact utility loaders have been difficult, non-intuitive, and burdensome to use.

(3) Often a compact utility loader will be maneuvered by traction elements (e.g., tracks or wheels) on either side of the loader. Generally, a compact utility loader will include a separate control element (e.g., a control handle) for controlling each traction element of the loader. The need for an operator to manipulate multiple control elements to maneuver a compact utility loader can make operation of the loader overly difficult and cumbersome. This is particularly true when the operator is required to maintain a free hand to control the loader's hydraulically-driven tools. As such, there is a need for an improved vehicle steering assembly, which enhances the ability of an operator to efficiently and intuitively control a vehicle, such as compact utility loader.

SUMMARY OF THE INVENTION

(4) In one embodiment of the present invention, there is provided a vehicle steering assembly for controlling movement of a vehicle having independently rotatable left and right ground-engaging traction elements. The vehicle comprises a control panel within reach of an operator of the vehicle.

The control panel comprises a panel support structure for supporting control mechanisms of the vehicle. The steering assembly comprises a T-shaped steering handle extending above the panel support structure, a steering controller positioned at least partly below the panel support structure, and a flexible protective cover attached to the steering handle. The steering handle comprises an upright shaft and a laterally extending crossmember rigidly coupled to the upright shaft. The upright shaft comprises a base that actuates the steering controller. The flexible protective cover extends around and is attached to the shaft. The flexible protective cover is configured to prevent dust and debris from entering the steering controller at the base of the shaft. The steering handle is shiftable relative to the panel support structure in forward and rearward directions to thereby cause corresponding forward and rearward rotation of both of the left and right traction elements. The steering handle is rotatable relative to the panel support structure in clockwise and counterclockwise directions to thereby cause a change in the relative speeds and directions of rotation of the left and right traction elements.

(5) In another embodiment of the present invention, there is provided a steering control assembly for controlling movement of a vehicle having left and right traction elements. The steering control assembly comprises a steering handle, a steering controller, a flexible protective cover, and an attachment ring. The steering handle comprises an upright shaft and a cross-member fixed to a top of the upright shaft. The cross-member is configured for manual grasping by an operator of the vehicle. The steering controller is actuated by the steering handle and is configured to control movement of the vehicle based on the positioning of the steering handle. The flexible protective cover is attached to the shaft and is configured to prevent dust and debris from entering the steering controller. The attachment ring attaches the flexible protective cover to the shaft. The attachment ring comprises an inner portion and an outer portion and the shaft defines a circumferential recess receiving the inner portion of the attachment ring. The flexible protective cover defines an interior recess receiving the outer portion of the attachment ring. The attachment ring acts as a bushing between the shaft and the flexible protective cover so that the shaft can be rotated without causing corresponding rotation of the flexible protective cover. The steering handle is biased toward a neutral position so that the steering handle automatically returns to the neutral position when not acted upon by the operator of the vehicle. The steering control assembly is configured such that (i) tilting the steering handle forward from the neutral position toward a forward position causes forward rotation of both the left and right traction elements, (ii) tilting the steering handle rearward from the neutral position toward a rearward position causes rearward rotation of both the left and right traction elements, (iii) twisting of the steering handle clockwise from the neutral position toward a clockwise twisted position causes forward rotation of the left traction element and rearward rotation of the right traction element, and (iv) twisting of the steering handle counterclockwise from the neutral position toward a counterclockwise twisted position causes forward rotation of the right traction element and rearward rotation of the left traction element.

(6) In another embodiment of the present invention, there is provided a method for controlling movement of a vehicle having independently rotatable left and right ground-engaging traction elements. The method includes a step of providing a steering handle comprising a laterally-extending crossmember and an upright extension member extending away from a control panel of the vehicle. The crossmember and the upright extension member are rigidly connected to one another so that shifting of the crossmember relative to the extension member is substantially prevented. An additional step includes shifting the steering handle in a forward direction to cause forward rotation of both of the left and right traction elements. An additional step includes shifting the steering handle in a rearward direction to cause rearward rotation of both of the left and right traction elements. A further step includes rotating the steering handle in a clockwise or counterclockwise direction to cause the left and right traction elements to rotate in opposing directions.

(7) This summary is provided to introduce a selection of concepts in a simplified form that are

further described below in the detailed description. This summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter. Other aspects and advantages of the present invention will be apparent from the following detailed description of the embodiments and the accompanying drawing figures.

Description

BRIEF DESCRIPTION OF THE FIGURES

- (1) Embodiments of the present invention are described herein with reference to the following drawing figures, wherein:
- (2) FIG. 1 is a left perspective view of a vehicle, in the form of a compact utility loader, with a steering control assembly according to embodiments of the present invention;
- (3) FIG. 2 is a top plan view of the compact utility loader of FIG. 1;
- (4) FIG. 3 is a top plan view of a control panel from the compact utility loader of FIGS. 1 and 2;
- (5) FIG. 4 is a rear elevation view of the control panel from FIG. 3;
- (6) FIG. 5 is an exploded view of a steering control assembly from the control panel from FIGS. 3 and 4;
- (7) FIG. 6 is a steering handle and a handle connection member from the steering control assembly from FIG. 5;
- (8) FIG. 7a is a top plan view of the steering control assembly from FIG. 5, particularly showing the steering control assembly in a neutral position;
- (9) FIG. 7b is side elevation view of the steering control assembly from FIG. 7a, particularly showing a vertical cross section of a steering handle, a handle connection member, and a portion of a steering controller;
- (10) FIG. 8a is a top plan view of the steering control assembly from FIG. 5, particularly showing the steering control assembly shifted in a forward position;
- (11) FIG. 8b is side elevation view of the steering control assembly from FIG. 8a, particularly showing a vertical cross section of a steering handle, a handle connection member, and a portion of a steering controller;
- (12) FIG. 9a is a top plan view of the steering control assembly from FIG. 5, particularly showing the steering control assembly shifted in a rearward position;
- (13) FIG. 9b is side elevation view of the steering control assembly from FIG. 9a, particularly showing a vertical cross section of a steering handle, a handle connection member, and a portion of a steering controller;
- (14) FIG. 10a is a top plan view of the steering control assembly from FIG. 5, particularly showing the steering control assembly rotated in a clockwise position;
- (15) FIG. 10b is side elevation view of the steering control assembly from FIG. 10a, particularly showing a vertical cross section of a steering handle, a handle connection member, and a portion of a steering controller;
- (16) FIG. 11a is a top plan view of the steering control assembly from FIG. 5, particularly showing the steering control assembly rotated in a counterclockwise position;
- (17) FIG. 11b is side elevation view of the steering control assembly from FIG. 11a, particularly showing a vertical cross section of a steering handle, a handle connection member, and a portion of a steering controller;
- (18) FIG. 12 is a vertical cross section of the steering control assembly from FIG. 5;
- (19) FIG. 13a is a top plan view of a handle connection member from the steering control assembly of FIG. 12;
- (20) FIG. 13b is a perspective view of the handle connection member from FIG. 13a;

(21) FIG. 13c is a side elevation view of the handle connection member from FIGS. 13a and 13b;

(22) FIG. 13d is an additional side elevation view of the handle connection member from FIGS. 13a, 13b, and 13c.

(23) FIG. 14 is a top plan view of a horizontal cross section of the steering control assembly from FIG. 12;

(24) FIG. 15 is a perspective view of the horizontal cross section of the steering control assembly from FIG. 14;

(25) FIG. 16 is a perspective view of a steering control assembly according to additional embodiments of the present invention;

(26) FIG. 17 is an exploded view of the steering control assembly from FIG. 16, with a portion of a steering handle cut away to show a handle connection assembly;

(27) FIG. 18 is an exploded view of the handle connection assembly from FIG. 17 in conjunction with a handle securement structure from the steering control assembly;

(28) FIG. 19 is a bottom perspective view of a pedestal from the handle connection assembly from FIG. 18;

(29) FIG. 20a is a top plan view of a horizontal cross section of the steering control assembly from FIG. 16;

(30) FIG. 20b is a perspective view of the horizontal cross section of the steering control assembly from FIG. 20a;

(31) FIG. 21 is a bottom perspective view of a steering handle and a handle connection assembly from the steering control assembly from FIG. 16;

(32) FIG. 22 is vertical cross section of the steering control assembly from FIG. 16;

(33) FIG. 23a is a top plan view of the steering control assembly from FIG. 16, particularly showing the steering control assembly in a neutral position;

(34) FIG. 23b is side elevation view of the steering control assembly from FIG. 23a, particularly showing a vertical cross section of a steering handle, a handle connection assembly, and a portion of a steering controller;

(35) FIG. 24a is a top plan view of the steering control assembly from FIG. 16, particularly showing the steering control assembly shifted in a forward position;

(36) FIG. 24b is side elevation view of the steering control assembly from FIG. 24a, particularly showing a vertical cross section of a steering handle, a handle connection assembly, and a portion of a steering controller;

(37) FIG. 25a is a top plan view of the steering control assembly from FIG. 26, particularly showing the steering control assembly shifted in a rearward position;

(38) FIG. 25b is side elevation view of the steering control assembly from FIG. 25a, particularly showing a vertical cross section of a steering handle, a handle connection assembly, and a portion of a steering controller;

(39) FIG. 26a is a top plan view of the steering control assembly from FIG. 16, particularly showing the steering control assembly rotated in a clockwise position;

(40) FIG. 26b is side elevation view of the steering control assembly from FIG. 26a, particularly showing a vertical cross section of a steering handle, a handle connection assembly, and a portion of a steering controller;

(41) FIG. 27a is a top plan view of the steering control assembly from FIG. 16, particularly showing the steering control assembly rotated in a counterclockwise position;

(42) FIG. 27b is side elevation view of the steering control assembly from FIG. 27a, particularly showing a vertical cross section of a steering handle, a handle connection assembly, and a portion of a steering controller;

(43) FIG. 28a is a side elevation schematic view of an additional embodiment of a steering control assembly in which a steering handle is configured to forwardly and rearwardly translated along a rail;

(44) FIG. **28b** is side elevation schematic view of the steering control assembly from FIG. **28a**;
(45) FIG. **29** is a vertical sectional view of an alternative steering control assembly employing an attachment ring to connect a flexible protective cover to the shaft of the steering handle;
(46) FIG. **30** is an exploded view of the steering control assembly from FIG. **29** and showing how the attachment ring can be connected to the shaft of the steering handle; and
(47) FIG. **31** is a partial cut-away view of the steering control assembly from FIGS. **29** and **30**.
(48) The drawing figures do not limit the present invention to the specific embodiments disclosed and described herein. The drawings are not necessarily to scale, emphasis instead being placed upon clearly illustrating the principles of the invention.

DETAILED DESCRIPTION

(49) The following detailed description of the present invention references various embodiments. The embodiments are intended to describe aspects of the invention in sufficient detail to enable those skilled in the art to practice the invention. Other embodiments can be utilized and changes can be made without departing from the scope of the present invention. The following detailed description is, therefore, not to be taken in a limiting sense. The scope of the present invention is defined only by the appended claims, along with the full scope of equivalents to which such claims are entitled.

(50) In this description, references to “one embodiment,” “an embodiment,” or “embodiments” mean that the feature or features being referred to are included in at least one embodiment of the technology. Separate references to “one embodiment,” “an embodiment,” or “embodiments” in this description do not necessarily refer to the same embodiment and are also not mutually exclusive unless so stated and/or except as will be readily apparent to those skilled in the art from the description. For example, a feature, structure, act, etc. described in one embodiment may also be included in other embodiments, but is not necessarily included. Thus, the present technology can include a variety of combinations and/or integrations of the embodiments described herein.

(51) Broadly, embodiments of the present invention are directed to a steering control assembly for a vehicle, such as for a compact utility loader. FIGS. **1** and **2** show a vehicle in the form of a compact utility loader **10** (hereinafter “loader **10**”) equipped with a steering control assembly **12(a)** that includes an improved steering handle **14(a)**. The steering handle **14(a)** controls movement of the loader **10**. Movement of the loader **10** is provided by independently rotatable left and right traction elements **16** (e.g., left and right endless tracks).

(52) As shown in FIGS. **1** and **2**, the loader **10** may include an operator platform **18** on which the operator stands while operating the loader **10**. The loader **10** has a control panel **20** within reach and view of the operator. The control panel **20** includes a panel support structure **22** to which the steering control assembly **12(a)** is connected. The loader **10** also includes a pair of lift arms **24** to which a working attachment **26** (e.g., a loader bucket) is connected.

(53) The loader **10** includes a power source (not shown), such as a diesel engine positioned below a hood **28** of the loader **10**, for powering the loader's **10** drive system and loader system. In certain embodiments, the power source can be a turbocharged diesel engine of less than 25 horsepower. Preferably, the power source has more than 20 horsepower. The drive system includes the loader's **10** drive train (e.g., hydraulic motors and/or a hydrostatic transmission) and the left and right traction elements **16** (e.g., tracks). The loader system includes the loader's **10** lift arms **24** and the working attachment **26** (e.g., loader bucket).

(54) Although FIGS. **1** and **2** show a compact utility loader **10** with traction elements **16** in the form of endless tracks, the loader **10** can be any type of vehicle having independently rotatable left and right traction elements **16** (e.g., tracks or wheels), such as, for example, a skid steer loader, an excavator, a tractor, or a bulldozer.

(55) Although FIGS. **1** and **2** show the working attachment **26** in the form of a loader bucket connected to the end of the lift arms **24**, the working attachment **26** could be selected from a variety of different useful tools, such as, for example, a trencher, a tiller, a posthole digger, a mower, a fork

lift, a grapple rake, a hydraulic breaker, a snow thrower, a box rake, a stump grinder, a utility blade, or a trench filler.

(56) Turning to FIG. 3 a top view of the loader's **10** control panel **20** is shown, illustrating that the control panel **20** can include an information portion, such as gauges, graphic displays, and the like, which provides visual information to the operator. The control panel **20** can additionally include a control portion, which can be manually manipulated by the operator to control the loader **10**. The control portion of the control panel **20** can include switches, the steering handle **14(a)**, a throttle handle **30**, and a loader handle **32**.

(57) As illustrated by the arrows in FIG. 3, the steering handle **14(a)**, the throttle handle **30**, and the loader handle **32** can be manually manipulated in various directions to control movement and speed of various portions of the loader **10**. In particular, the steering handle **14(a)** can be manually manipulated forwardly and rearwardly and can be manually twisted in clockwise and counterclockwise directions to control the speed and direction of rotation of the left and right traction elements **16**. As used herein, the terms “forward” or “forwardly” refer to a direction towards a front of the loader **10**, such as where the working attachment **26** is connected to the loader **10**. The terms “rearward” or “rearwardly” refer to a direction towards a back of the loader **10**, such as where the operator stands on the operator platform **18** to operate the loader **10**. The terms “right” and “left” means a right direction and a left direction, respectively, when viewing forwardly from the back of the loader **10**. The terms “clockwise” and “counterclockwise” means a clockwise direction and a counterclockwise direction, respectively, when looking down on the applicable component from above the component. Thus, with respect to the view of the control panel **20** on FIG. 2, a forward direction is toward the left side of the figure, a rearward direction is toward the right side of the figure, a left direction is toward the bottom side of the figure, a right direction is toward the top side of the figure, a clockwise direction is clockwise (as looking down on the loader **10**), and a counter-clockwise direction is counter-clockwise (as looking down on the loader **10**).

(58) FIG. 4 is a vertical view from the rear of the panel support structure **22** of the control panel **20**, with the panel support structure shown supporting the steering control assembly **12(a)**, a throttle control assembly **40**, and a lift control assembly **42**. The steering, throttle, and lift control assemblies **12(a)**, **40**, and **42** include manually movable handles (i.e., steering handle **14(a)**, throttle handle **30**, and lift handle **32**, respectively) each being connected to a controller. The handles **14(a)**, **30**, **32** extending generally above the panel support structure **22** and the corresponding controllers extending generally below the panel support structure **22**. Each controller is rigidly coupled to the panel support structure **22** and each handle **14(a)**, **30**, **32** is shiftable relative to the controller to which it is coupled. In some embodiments, the controllers may be secured to the panel support structure **22** in a manner that permits an upper portion of the controllers (e.g., depressible buttons disclosed in more detail below) to extend upward above the panel support structure **22**. Although not shown in the drawing, a protective cover **33** (e.g., flexible boot) can be attached to the shaft **35** of each handle **14(a)**, **30**, **32** to prevent dust and debris from entering the controllers at the base **54** of the shaft **35**.

(59) FIG. 5 is an exploded view of the steering control assembly **12(a)**. As shown in FIG. 5, the controller of the steering control assembly **12(a)** is a steering controller **44(a)**, which in certain embodiments may be a hydraulic pilot control valve in communication with the loader's **10** drive train (e.g., hydraulic motors and/or a hydrostatic transmission). The steering controller **44(a)** is equipped with four depressible buttons **46** that cooperatively control the speed and direction of rotation of the loader's **10** traction elements **16** (e.g., tracks or wheels) via communication between the steering controller **44(a)** and the loader's **10** drive train (e.g., hydraulic motors and/or a hydrostatic transmission). A front left button **46** may control the forward direction of the left the traction element **16**, while a rear left button **46** may control the rearward direction of the left the traction element **16**. In some embodiments, the rear left button **46** will be position behind the front

left button **46**. Similarly, a front right button **46** may control the forward direction of the right the traction element **16**, while a rear right button **46** may control the rearward direction of the right the traction element **16**. In some embodiments, the rear right button **46** may be position behind the front right button **46**. In addition, the front right and the rear right buttons **46** may be positioned, respectively, to the right of the front left and the rear left buttons **46**. The steering handle **14(a)** is coupled to the steering controller **44(a)** and is used to depress the buttons **46** in a manner that, as will be described in more detail below, allows for simple and intuitive control of the movement of the loader.

(60) Remaining with FIG. **5**, the steering control assembly **12(a)** includes a handle securement structure **50(a)** that is used to connect the steering handle **14(a)** to the steering controller **44(a)** in a manner that allows the steering handle **14(a)** to be shifted forward, shifted rearward, rotated clockwise, and rotated counterclockwise relative to the steering controller **44(a)**, as described in more detail below. In some embodiments, the handle securement structure **50(a)** will be rigidly secured to the steering controller **44(a)** via a plurality of threaded fasteners that extend down through both the handle securement structure **50(a)** and the steering controller **44(a)**. The steering controller **44(a)** may itself be rigidly secured to the panel support structure **22** via a plurality of threaded fasteners that extend down through both the panel support structure **22** and the steering controller **44(a)**.

(61) The steering handle **14(a)** may include an upper crossmember **52**, a lower base **54**, and an upright extension member **56** extending between the base **54** and the crossmember **52**. The crossmember **52** and the upright extension member **56** can be connected in a generally T-shaped configuration. In some embodiments, the crossmember **52** and the upright extension member **56** may be rigidly connected to one another so that shifting of the crossmember **52** relative to the upright extension member **56** is substantially (or completely) prevented. Furthermore, in some embodiments, the steering control assembly **12(a)** will be connected to the panel support structure **22** in such a manner that the crossmember **52** is substantially (or completely) prevented from lateral shifting relative to the panel support structure **22**. In such embodiments, for instance, only forward, rearward, and rotating movements of the steering handle **14(a)** relative to the panel support structure **22** will be permitted.

(62) A bottom of the base **54** may presents a curved lower or bottom surface **57** that is configured to contact the depressible buttons **46** and selectively depress the buttons **46** as the steering handle **14(a)** is manually manipulated. In some of such embodiments, the crossmember **52**, the upright extension member **56**, and the base **54** will be integrally formed from a unitary piece of material.

(63) Although the crossmember **52** of the steering handle **14(a)** is depicted herein as being a substantially straight elongated member that is connected to the top of the upright extension member **56** to form a T-shape configuration, it should be understood that the crossmember **52** can take on a variety of shapes for facilitating manual grasping by the operator of the loader **10**. For example, the crossmember **52** can have an irregular ergonomic shape that conforms to the hand of the operator. In certain specific embodiments, the steering handle **14(a)** will only include a single upright extension member **56**.

(64) FIG. **6** shows the bottom of the steering handle **14(a)** in more detail. In particular, FIG. **6** shows that the curved bottom surface **57** of the base **54** includes a curved topography with a substantially flat front section **60**, a substantially flat rear section **62**, a pair of left-side downwardly sloping sections **64**, and a pair of right-side downwardly sloping sections **66**. The left-side downwardly sloping sections **64** are formed on the sides of a left downward projection **68** of the base **54**. It should be understood that the left downward projection **68** extends down below the front and rear sections **60**, **62**. The right-side downwardly sloping sections **66** are formed on the sides of a right downward projection **69** of the base **54**. It should be understood that the right downward projection **69** extends down below the front and rear sections **60**, **62**.

(65) FIG. **6**, also shows that the base **54** includes an internal opening within which a handle

connection member **70(a)** is received. The handle connection member **70(a)** is used to connect the steering handle **14(a)** with the steering controller **44(a)** in a manner that allows the steering handle **14(a)** to be shifted forward, shifted rearward, rotated clockwise, and rotated counterclockwise relative to the steering controller **44(a)**, as described in more detail below.

(66) FIGS. **7a** (top view) and **7b** (side view) show the steering control assembly **12(a)** in a neutral position. When the steering control assembly **12(a)** is in the neutral position, none of the depressible buttons **46** are depressed enough to cause rotation of the right or left traction elements **16** of the loader **10**. The steering control assembly **12(a)** is biased toward this neutral position so that if the operator of the loader **10** releases a hand grip on the steering handle **14(a)**, the loader **10** stops.

(67) As illustrated by FIG. **7b**, the steering controller **44(a)** includes, for each depressible button **46**, a variable switch **71** (or pilot valve spool) and a biasing mechanism **72** for biasing the depressible buttons **46** upwardly. The variable switch **71** may comprise a position sensor that measures the depth of depression of the depressible button **46** and permits the speed of rotation of the traction elements **16** to be controlled in a manner that is proportional to the depth of depression of the depressible button **46**. In some embodiments, as noted above, the steering controller **44(a)** may be a hydraulic pilot control valve and movement of the depressible buttons **46** directly adjusts the flow or pressure of hydraulic fluid through the control valve. For example, each of the depressible buttons **46** may activate an associated variable switch **71** and/or may move an associated spool in the steering controller **44(a)** when the depressible button **46** is depressed. Such actuation of the variable switch **71** and/or the spool will generate (or change) a pilot pressure signal that is sent from the steering controller **44(a)** to the loader's **10** drive train (e.g., the hydrostatic transmission that controls the hydraulic motors). The pilot pressure signal sent to the loader's **10** drive train will be based on the depth at which the depressible button **46** is depressed. As such, when the depressible button **46** is fully depressed, a pilot pressure signal is sent to the loader's **10** drive train (e.g., to the hydrostatic transmission) to cause the associated traction element **16** to be rotated at a maximum rate. If the depressible button **46** is depressed a lesser amount (i.e., less than a full depression), a pilot pressure signal is sent to the loader's **10** drive train (e.g., to the hydrostatic transmission) to cause the associated traction element **16** to be rotated at a rate that is less than the maximum rate. It is further understood that one depressible button **46** (e.g., the front left button **46**) is used to generate a pilot control signal that causes the left traction element **16** to rotate in a forward direction, while another depressible button **46** (e.g., the rear left button **46**) is used to generate a pilot control signal that causes the left traction element **16** to rotate in a rearward direction. Similarly, one depressible button **46** (e.g., the front right button **46**) is used to generate a pilot control signal that causes the right traction element **16** to rotate in a forward direction, while another depressible button **46** (e.g., the rear right button **46**) is used to generate a pilot control signal that causes the right traction element **16** to rotate in a rearward direction.

(68) The biasing mechanism **72** associated with each depressible button **46** pushes the button **46** up against the bottom surface **57** of the steering handle **14(a)**. When the steering handle **14(a)** is not being manually manipulated out of the neutral position, all the depressible buttons are fully extended and the steering handle **14(a)** is maintained in the neutral position by the depressible buttons **46** pushing up against the substantially flat front and rear sections **60**, **62** of the curved bottom surface **57** of the base **54**.

(69) FIGS. **7a** and **7b** show that the steering handle **14(a)** is connected to the steering controller **44(a)** in a manner that allows the steering handle **14(a)** to be shifted/tilted forward and rearward on a tilt axis **74**. Further, the steering handle **14(a)** is connected to the steering controller **44(a)** in a manner that allows the steering handle **14(a)** to be rotated/twisted clockwise and counterclockwise on a twist axis **76**. The tilt axis **74** and twist axis **76** are maintained substantially perpendicular to one another, even during shifting or rotating of the steering handle **14(a)**.

(70) As shown in FIGS. **8a** (top view) and **8b** (side view) shifting the steering handle **14(a)** forward

on the tilt axis **74** (See FIG. **8b**), without rotating the steering handle **14(a)** on the twist axis **76** (See FIG. **8b**), depresses the front left and right depressible buttons **46**. When the front left and right depressible buttons **46** are depressed an equal amount, as shown in FIGS. **8a** and **8b**, the left and right traction elements **16** rotate forward at substantially the same speed, so the loader **10** travels straight forward. It should be understood that the further the front left and right depressible buttons **46** are depressed the faster the left and right traction elements **16** will rotate so as to increase the speed of the loader **10**. As such, when the steering handle **14(a)** is shifted fully forward, the left and right front depressible buttons **46** will be depressed a maximum amount, and the left and right traction elements **16** will rotate forwardly at a maximum rate to cause the loader to drive forward at a maximum speed.

(71) As shown in FIGS. **9a** (top view) and **9b** (side view), shifting the steering handle **14(a)** rearward on the tilt axis **74**, without rotating the steering handle **14(a)** on the twist axis **76**, depresses the rear left and right depressible buttons **46**. When the rear left and right depressible **46** buttons are depressed an equal amount, as shown in FIGS. **9a** and **9b**, the left and right traction elements **16** rotate rearward at substantially the same speed, so the loader **10** travels straight backward (i.e., in reverse). It should be understood that the further the rear left and right depressible buttons **46** are depressed the faster the left and right traction elements **16** will rotate so as to increase the speed of the loader **10**. As such, when the steering handle **14(a)** is shifted fully rearward, the left and right rear depressible buttons **46** will be depressed a maximum amount, and the left and right traction elements **16** will rotate rearwardly at a maximum rate to cause the loader to drive backward at a maximum speed.

(72) As shown in FIGS. **10a** (top view) and **10b** (side view), rotating the steering handle **14(a)** clockwise on the twist axis **76**, without shifting the steering handle **14(a)** forward or rearward on the tilt axis **74**, depresses the front left and rear right depressible buttons **46**. When the front left and rear right depressible buttons **46** are depressed an equal amount and the front right and rear left depressible buttons **46** are not depressed, as shown in FIGS. **10a** and **10b**, the left and right traction elements **16** rotate at substantially the same speed, with the left traction element **16** rotating forward and the right traction element **16** rotating rearward. In this configuration, the loader **10** turns clockwise without traveling forward or backward. It should be understood that the further the front left and rear right depressible buttons **46** are depressed the faster the left and right traction elements **16** will rotate so as to increase the turning speed or severity of the turn (i.e., the right turn) of the loader **10**. As such, when the steering handle **14(a)** is rotated clockwise a full amount forward, the front left and rear right depressible buttons **46** will be depressed a maximum amount, and the left and right traction elements **16** will rotate oppositely at a maximum rate to cause the loader to turn in a clockwise direction at a maximum speed/severity.

(73) As shown in FIGS. **11a** (top view) and **11b** (side view), rotating the steering handle **14(a)** counterclockwise on the twist axis **76**, without shifting the steering handle **14(a)** forward or rearward on the tilt axis **74**, depresses the front right and rear left depressible buttons **46**. When the front right and rear left depressible buttons **46** are depressed an equal amount and the front left and rear right depressible buttons **46** are not depressed, as shown in FIGS. **11a** and **11b**, the right and left traction elements **16** rotate at substantially the same speed, with the right traction element **16** rotating forward and the left traction element **16** rotating rearward. In this configuration, the loader **10** turns counterclockwise without traveling forward or backward. It should be understood that the further the front right and rear left depressible buttons **46** are depressed the faster the left and right traction elements **16** will rotate so as to increase the turning speed or severity of the turn (i.e., the left turn) of the loader **10**. As such, when the steering handle **14(a)** is rotated counterclockwise a full amount forward, the rear left and front right depressible buttons **46** will be depressed a maximum amount, and the left and right traction elements **16** will rotate oppositely at a maximum rate to cause the loader to turn in a counterclockwise direction at a maximum speed/severity.

(74) When the steering handle **14(a)** is simultaneously tilted (forward or rearward) and twisted

(clockwise or counterclockwise), both the speed (fast or slow) and direction of travel (forward, backward, and turning) of the loader **10** are simple and intuitively controlled by the operator using a single one hand on the steering handle **14(a)**.

(75) FIG. **12** is a sectional view of the steering handle **14(a)** and steering controller **44(a)** in the neutral position. FIG. **12** also shows how the steering handle **14(a)** is coupled to the steering controller **44(a)** by a tilt hinge **84** and a twist hinge **86**. The tilt hinge **86** is configured to permit the steering handle **14(a)** to rotate relative to the steering controller **44(a)** (and the panel support structure **22**) on the tilt axis **74** (See FIG. **7b**), while the twist hinge **86** is configured to permit the steering handle **14(a)** to rotate relative to the steering controller **44(a)** (and panel support structure **22**) on the twist axis **76** (See FIG. **7b**).

(76) The tilt hinge **84** can be formed by the handle securement structure **50(a)**, the handle connection member **70(a)**, and a hinge pin **88**. More specifically, the handle securement structure **50(a)** is rigidly coupled to the steering controller **44(a)**, the handle connection member **70(a)** is connected to the steering handle **14(a)**, and the hinge pin **88** extends through openings in both the handle connection member **70(a)** and the handle securement structure **50(a)** to secure the components together in a manner that permits the tilting of the handle connection member **70(a)** relative to the handle securement structure **50(a)** on the tilt axis **74**. Generally, the hinge pin **88** is aligned with the tilt axis **74**.

(77) The twist hinge **76** can be formed by the handle connection member **70(a)** and the base **54** of the steering handle **14(a)**. More specifically, the handle connection member **70(a)** includes a head **90** that is received in an internal cavity of the base **54** in a manner that permits the steering handle **14(a)** to rotate relative to the head **90** and the handle connection member **70(a)** on the twist axis **76**. Generally, the head **90** and/or the handle connection member **70(a)** are aligned with the twist axis **76**. Correspondingly, the upright extension member **56** of the steering handle **14(a)** will generally be aligned with and/or centered on the twist axis **76**, while the crossmember **52** is perpendicular to the twist axis **76** with the twist axis **76** extending through the crossmember **52**.

(78) As such, the steering control assembly **12(a)** may comprises a handle connection assembly for attaching the steering handle **14(a)** to the panel support structure **22**, whereby said handle connection assembly comprises the tilt hinge **84** and the twist hinge **86**. The tilt hinge **84** is configured to permit the steering handle **14(a)** to tilt forward and rearward relative to the panel support structure **22** on a tilt axis **74**, while the twist hinge **86** is configured to permit the steering handle **14(a)** to twist clockwise and counterclockwise relative to the panel support structure **22** on a twist axis **76**.

(79) FIGS. **13a-d** provides various views of the handle connection member **70(a)** that can be received within the internal cavity of the steering handle **14(a)**. Specifically, FIGS. **13a-d** show that the handle connection member **70(a)** can includes an extension member **92**, a neck **94**, the head **90**, and a projection **96**.

(80) The extension member **92** includes an attachment opening **98** through which the hinge pin **84** can extend, so as to couple the handle connection member **70(a)** to the steering controller **44(a)** (and the panel support structure **22**) via the handle securement structure **50(a)** in a manner that permits pivoting of the handle connection member **70(a)** relative to the panel support structure **50(a)** on the tilt axis **74**. The head **90** is configured to be received in a broad portion of the internal cavity of the steering handle **14(a)**, as illustrated in FIG. **12**. The neck **94** is configured to be received in a narrow portion of the internal cavity in a manner that prevents the head **90** from being pulled out of the internal cavity but permits rotation of the steering handle **14(a)** relative to the handle connection member **70(a)** on the twist axis **76**. The projection **96** of the handle connection member **70(a)** extends upwardly from the head **90** and is received in an upper portion of the internal cavity of the steering handle **14(a)**.

(81) FIG. **14** is a horizontal cross-sectional view cut through the steering handle **14(a)** at the very top of the internal cavity that receives the head **90** of the handle connection member **70(a)**. FIG. **14**

shows that the steering handle **14(a)** includes stops **100** that extend into the broad portion of the internal cavity at the same level as the projection **92** of the handle connection member **70(a)**. These stops **100** may form part of a twist stop assembly, which are configured to be positioned within the steering handle **14(a)** to restrict rotation of the steering handle **14(a)** on the twist axis **76** within a certain range. When the steering handle **14(a)** is fully twisted in a clockwise or counterclockwise manner, stop surfaces **102** of the projection **92** contact the stops **100** and prevent further rotation of steering handle **14(a)** relative to the handle connection member **70(a)**, steering controller **44(a)**, and panel support structure **22**.

(82) In some embodiments, the steering handle **14(a)** is rotatable on the twist axis **76** through a twist range of motion that is between 15 and 80 degrees, or 25 and 60 degrees, or 30 and 50 degrees. In addition, the steering handle **14(a)** is shiftable on the tilt axis through a tilt range of motion that is between 15 and 80 degrees, or 25 and 60 degrees, or 30 and 50 degrees.

(83) Stating the above differently, in some embodiments, the handle connection assembly comprises a panel attachment piece (e.g., the handle securement structure and/or upper portions of the steering controller **44(a)**), a handle attachment piece (e.g., handle connection member **70(a)**), and a tilt hinge pin **88**. The panel attachment piece can be attached to the panel support structure **22**. The handle attachment piece can be attached to the handle **14(a)**. And the steering handle **14(a)** and the panel attachment pieces are coupled to one another by the tilt hinge pin **88**, with the tilt hinge pin **88** extending along the tilt axis **74**. In some embodiments, the panel attachment piece will be rigidly coupled to the panel support structure **22**, and the handle attachment piece will be rotatably coupled to the steering handle **14(a)**. The handle attachment piece includes a upper broadened head. The steering handle **14(a)** defines an internal head-receiving cavity within which said broadened head is received to thereby couple the handle attachment piece to the steering handle **14(a)**. As a result, the steering handle is rotatable relative to the broadened head on the twist axis **76**. Finally, the broadened head includes at least one projection, and the steering handle **14(a)** includes at least one stop member. Contact between the projection and the stop member restricts rotation of the steering handle **14(a)** relative to the panel support structure **22** on said twist axis **76**.

(84) As shown in FIGS. **14** and **15**, as well as FIG. **7b**, the steering handle **14(a)** can be formed of two halves—a first handle half **14(a)(i)** and a second handle half **14(a)(ii)**. With the steering handle **14(a)** in the neutral, upright position, the first handle half **14(a)(i)** may be rearward of the second handle half **14(a)(ii)** and separated by a vertical plane. The first handle half **14(a)(i)** may be secured to the second handle half **14(a)(ii)** via one or more forward/rearward extending fasteners. In some embodiments, the first handle half **14(a)(i)** and the second handle half **14(a)(ii)** may be mirror images of each other. In some specific embodiments, at least interior portions of the first handle half **14(a)(i)** and the second handle half **14(a)(ii)** (e.g., the internal cavity) may be mirror images of each other. In some embodiments, the crossmember **52**, the upright extension member **56**, and the base **54** of the first handle half **14(a)(i)** may be integrally formed from a unitary piece of material, while the crossmember **52**, the upright extension member **56**, and the base **54** of the second handle half **14(a)(ii)** may be integrally formed from a unitary piece of material.

(85) Forming the handle **14(a)** in two halves, allows for easy attachment of the handle connection member **70(a)** to the steering handle **14(a)** by (i) placing the head **90** of the handle connection member **70(a)** in the portion of the internal chamber defined by the first handle half **14(a)(i)**, (ii) aligning the second handle half **14(a)(ii)** with the first handle half **14(a)(i)** so that the head is the portion of the internal chamber defined by the second handle half **14(a)(ii)**, and (iii) coupling the two handle halves **14(a)(i)** and **14(a)(ii)** to one another (e.g., via the fasteners) while the head **90** is received in the internal cavity that is cooperatively formed by the first handle half **14(a)(i)** and the second handle half **14(a)(ii)**.

(86) Another embodiment of a steering control assembly **12(b)** is illustrated in FIG. **16**. Broadly, the steering control assembly **12(b)** will include a steering handle **14(b)** secured to a steering controller **44(a)**. The steering control assembly **12(b)** may include many of the same components

and may be configured to perform many of the same functions as the steering control assembly **12(a)** discussed above. As with steering control assembly **12(a)**, the steering control assembly **12(b)** is configured such that steering handle **14(b)** can be manually manipulated forwardly and rearwardly and can be manually twisted in clockwise and counterclockwise directions with respect to the steering controller **44(b)** to selectively depress buttons **46** to control the speed and direction of rotation of the left and right traction elements **16** of the loader **10**.

(87) As shown in FIG. **17**, the steering control assembly **12(b)** comprises a handle connection assembly **110** that is used to secure the steering handle **12(b)** to the steering controller **44(b)**. The handle connection assembly **110** may include a handle connection member **70(b)** configured to extend upward into the internal cavity of the steering handle **14(b)** in a similar manner as the handle connection member **70(a)**. However, the handle connection member **70(b)** may be significantly longer than the handle connection member **70(a)**, such that the handle connection member **70(b)** may extend into the internal cavity of the steering handle **14(b)** up through base **54**, through the upright extension member **56** and to (or into) the crossmember **52**. The handle connection member **70(b)** may include a head **112** at its upper end and an annular base **114** at its lower end.

(88) The handle connection assembly **110** may additionally include a pedestal **120** to which the handle connection member **70(b)** may be rigidly secured. In particular, the annular base **114** of the handle connection member **70(b)** may be secured to an upper surface of the pedestal **120** via fasteners. Notably, a bottom of the pedestal may be formed with a curved bottom surface **122**, which is formed similarly to the curved bottom surface **57** of the base **54** of the steering handle **14(a)** discussed above. Specifically, the curved bottom surface **122** may include a substantially flat front section, a substantially flat rear section, a pair of left-side downwardly sloping sections, and a pair of right-side downwardly sloping sections. As such, the left-side downwardly sloping sections are formed on the sides of a left downward projection of the pedestal **120**, and the right-side downwardly sloping sections are formed on the sides of a right downward projection of the pedestal **120**. As a result, the curved bottom surface **122** of the steering assembly **12(b)** is formed on the handle connection assembly **110** that secures the steering handle **14(b)** to the steering controller **44(b)**.

(89) The handle connection assembly **110** may additionally include a tilt member **124**, which is configured to secure the handle connection member **70(b)** and the pedestal **120** to the steering controller **44(b)**. In particular as perhaps best illustrated by FIG. **18**, The tilt member **124** may be secured to the pedestal **120** via a threaded fastener **126** and associated washer, which permit the pedestal **120** to rotate about a twist axis **76** with respect to the tilt member **124**. As shown, the twist axis **76** will generally be aligned longitudinally with the threaded fastener **126**, the tilt member **124**, and the handle connection member **70(b)**. The tilt member **124** will be secured to a second embodiment of a handle securement structure **50(b)** via a pivot pin **128**. The pivot pin **128** will be held in place within the tilt member **124** via a set screw **129** extending through the tilt member **124** and into engagement with the pivot pin **128**. With the pivot pin **128** securing the tilt member **124** to the handle securement structure **50(b)**, the pivot pin **128** will be aligned (and will present) the tilt axis **74**. As illustrated in FIG. **17**, the handle securement structure **50(b)** is secured to the steering controller **40(b)**, such that the handle connection assembly **110** and the steering handle **14(b)** are operably secured to the steering controller **44(b)** (e.g., to depress associated buttons **46** as required).

(90) Returning to the pedestal **120**, as perhaps best shown by FIG. **19**, the pedestal includes an internal cavity with a pair of inwardly-orientated, triangular-shaped stops **130**, which may form part of a twist stop assembly. When the pedestal **120** is secured to the tilt member **124**, the stops **130** will be positioned at the same level as stop surfaces **132** extending from the tilt member **124** (See stop surfaces **132** on FIG. **18**). FIGS. **20(a)** and **20(b)** show a horizontal cross-sectional view cut through the pedestal **120** at the internal cavity that receives tilt member **124**. The figures show that the pedestal **120** includes the stops **130** extending into the internal cavity at the same level as

the stop surfaces **132** of the tilt member **124**. These stops **130** are configured to restrict rotation of the steering handle **14(b)** on the twist axis **76** within a certain range. When the steering handle **14(b)** is fully twisted in a clockwise or counterclockwise manner, stop surfaces **132** of the tilt member **124** contact the stops **130** and prevent further rotation of steering handle **14(b)** relative to the handle connection assembly **110** (including the handle connection member **70(b)**), the steering controller **44(b)**, and panel support structure **22**.

(91) In some embodiments, the steering handle **14(b)** is rotatable on the twist axis **76** through a twist range of motion that is between 15 and 80 degrees, or 25 and 60 degrees, or 30 and 50 degrees. In addition, the steering handle **14(b)** is shiftable on the tilt axis **74** through a tilt range of motion that is between 15 and 80 degrees, or 25 and 60 degrees, or 30 and 50 degrees.

(92) FIG. **21** further illustrates the steering handle **14(b)** coupled together with the handle connection assembly **110**. FIG. **22** is a vertical cross-section showing how the components of the steering control assembly **12(b)** fit together. FIGS. **23(a)-27(b)** show how the steering handle **14(b)** can be manipulated with respect to the steering controller **44(b)** to control the loader **10**.

(93) FIGS. **23a** (top view) and **23b** (side view) show the steering control assembly **12(b)** in a neutral position. When the steering control assembly **12(b)** is in the neutral position, none of the depressible buttons **46** are depressed enough to cause rotation of the right or left traction elements **16** of the loader **10**. The steering control assembly **12(b)** is biased toward this neutral position so that if the operator of the loader **10** releases a hand grip on the steering handle **14(b)**, the loader **10** stops. Specifically, the steering controller **44(b)** can include the variable switch and a biasing mechanism for biasing the depressible buttons **46** upwardly, similar to that described for steering controller **44(a)** above. When the steering handle **14(a)** is not being manually manipulated out of the neutral position, all the depressible buttons **46** are fully extended and the steering handle **14(b)** is maintained in the neutral position by the depressible buttons **46** pushing up against the substantially flat front and rear portions of the curved bottom surface **122** of the pedestal **120**. In some embodiments, the steering controller **44(b)** may be a hydraulic pilot control valve and movement of the depressible buttons **46** directly adjusts the flow of hydraulic fluid through the control valve to control direction and speed of the right and left traction elements **16** via the hydraulic motors.

(94) FIGS. **23a** and **23b** show that the steering handle **14(b)** is connected to the steering controller **44(b)** in a manner that allows the steering handle **14(b)** to be shifted/tilted forward and rearward on tilt axis **74**. Further, the steering handle **14(b)** is connected to the steering controller **44(b)** in a manner that allows the steering handle **14(b)** to be rotated/twisted clockwise and counterclockwise on twist axis **76**. The tilt axis **74** and twist axis **76** are maintained substantially perpendicular to one another, even during shifting or rotating of the steering handle **14(b)**.

(95) Certain of the remaining functionality of the steering control assembly **12(b)** is similar to that described above for steering control assembly **12(a)**. As shown in FIGS. **24a** (top view) and **24b** (side view) steering handle **14(b)** can be tilted forward such that the right and left traction elements **16** rotate forward at substantially the same speed, so the loader **10** travels straight forward. As shown in FIGS. **25a** (top view) and **25b** (side view), the steering handle **14(b)** can be shifted rearward causing the right and left traction elements **16** rotate rearward at substantially the same speed, so the loader **10** travels straight backward (i.e., in reverse). As shown in FIGS. **26a** (top view) and **26b** (side view), the steering handle **14(b)** clockwise such that the right and left traction elements **16** rotate at substantially the same speed, with the left traction element **16** rotating forward and the right traction element **16** rotating rearward. In this configuration, the loader **10** turns clockwise without traveling forward or backward. Finally, as shown in FIGS. **27a** (top view) and **27b** (side view), the steering handle **14(b)** can be rotated counterclockwise such that the right and left traction elements **16** rotate at substantially the same speed, with the right traction element **16** rotating forward and the left traction element **16** rotating rearward. In this configuration, the loader **10** turns counterclockwise without traveling forward or backward.

(96) When the steering handle **14(b)** is simultaneously tilted (forward or rearward) and twisted (clockwise or counterclockwise), both the speed (fast or slow) and direction of travel (forward, backward, and turning) of the vehicle are easily and intuitively controlled by the operator using a single one hand on the steering handle **14(b)**.

(97) As shown in FIG. 17, the steering handle **14(b)** can be formed of two halves—a first handle half **14(b)(i)** and a second handle half **14(b)(ii)**. The first handle half **14(a)(i)** may be secured to the second handle half **14(a)(ii)** via one or more forward/rearward extending fasteners or a snap fit-connection. Forming the handle **14(b)** in two halves, allows for easy attachment of the handle connection assembly **110**, including the handle connection member **70(b)** to the steering handle **14(b)** by (i) placing the handle connection member **70(b)** in the internal chamber defined by the first handle half **14(b)(i)**, (ii) aligning the second handle half **14(b)(ii)** with the first handle half **14(b)(i)** so that the handle connection member **70(b)** is the portion of the internal chamber defined by the second handle half **14(b)(ii)**, and (iii) coupling the two handle halves **14(b)(i)** and **14(b)(ii)** to one another (e.g., via the fasteners) while the handle connection member **70(b)** is received in the internal cavity that is cooperatively formed by the first handle half **14(b)(i)** and the second handle half **14(b)(ii)**.

(98) Although the steering handle assemblies **12(a)**, **12(b)** described above are configured to pivot/rotate on two axes (i.e., a tilt axis **74** and the twist axis **76**), it should be understood that other embodiments of a steering handle assembly may be configured to include a steering handle that can translate forward, rearward, and/or side-to-side relative to a steering controller, rather than pivot/rotate relative to the steering controller. For example, in the embodiment depicted FIGS. **28a** and **28b**, an embodiment of a steering handle assembly **12(c)** is illustrated with a steering handle **14(c)** mounted on a rail **150** that permits forward and rearward translation of the steering handle **14(c)** relative to a steering controller (identified by the depressible buttons **46**). In such a configuration, forward and rearward movement of the loader **10** can be controlled by sliding the steering handle **14(c)** forward and rearward on the rail **150**, while turning of the loader **10** can be controlled by tilting the steering handle **14(c)** left and right on an axis that is coextensive with the rail **150**.

(99) FIGS. **29-31** illustrate an alternative embodiment of a steering control assembly **12(d)** having an attachment ring **160** that is used to connect an upper end of the flexible protective cover **33** (e.g., a rubber boot) to the shaft **35** of the T-shaped steering handle **14(d)**. As mentioned above, the protective cover **33** can be used to prevent dirt and debris from entering the steering controller **44(d)** via an opening **23** in the panel support structure **22** through which the steering control assembly **12(d)** extends.

(100) In addition to functioning as an attachment mechanism for connecting the upper portion of the flexible protective cover **33** to the shaft **35**, the attachment ring **160** can, optionally, function as a bushing that allows for rotation of the shaft **35** within the protective cover **33**, so that the shaft **35** can be twisted clockwise and counterclockwise without requiring corresponding rotation and deformation of the protective cover **33**.

(101) When the attachment ring **160** acts as a bushing between the shaft **35** and the protective cover **33**, wear on the protective cover **33** caused by repeated twisting/deformation of the protective cover **33** can be minimized. This bushing feature of the attachment ring **160** can also facilitate smoother and easier twisting of the steering handle **14(d)** because the protective cover **33** provides little or no resistance to twisting the steering handle **14(d)**; rather, the steering handle **14(d)** simply twists relative to the protective cover **33** without causing substantial deformation of the protective cover **33**. Further, use of the attachment ring **160** can help facilitate the return of the steering handle **14(d)** to the neutral position after the steering handle **14(d)** has been twisted, thereby ensuring that the steering handle **14(d)** does not get stuck in a non-neutral, twisted position.

(102) To facilitate its use as a bushing, the attachment ring **160** can be formed of a low friction material. Such low friction material can having a coefficient of static friction of less than 1.0, less

than 0.5, less than 0.3, less than 0.2, or less than 0.1 and/or a coefficient of dynamic friction of less than 0.5, less than 0.4, less than 0.3, less than 0.2, less than 0.15, less than 0.1, or less than 0.05. The foregoing coefficient of friction values are measured using ASTM D1894. Additionally, it may be necessary for the attachment ring **160** to be formed of a flexible, resilient material that permits expansion of the attachment ring **160** in order to increase the inner diameter of the attachment ring **160** during installation.

(103) In certain embodiments, the attachment ring **160** can be at least partially formed of a polyamide and/or a fluoropolymer material. The polyamide can be nylon and the fluoropolymer can be polytetrafluoroethylene (PTFE). In other embodiments, the attachment ring **160** is formed predominately (>50% by weight) of nylon. In still other embodiments, the attachment ring **160** is formed predominately (>50% by weight) of PTFE.

(104) As perhaps best illustrated in FIG. **30**, the attachment ring **160** includes an inner portion **162** for engaging the shaft **35** of the steering handle **14(d)** and an outer portion **164** for engaging the protective cover **33**. In one embodiment, the attachment ring **160** includes a slit **164** that can be widened/opened by deforming/expanding the attachment ring **160**, thereby allowing the attachment ring to be install around the shaft **35**.

(105) As perhaps best shown in FIGS. **29** and **30**, the shaft **35** can have a circumferential recess **168** formed therein for receiving the inner portion **162** of the attachment ring **160**, thereby preventing the attachment ring **160** from sliding up and down the shaft **35**. FIG. **30** shows several options (“A”-“D”) for how the attachment ring **160** can be connected to the shaft **35** of the steering handle **14(d)** at the circumferential recess **168**. Option “A” includes the following steps: (a) sliding the attachment ring **160** over the handle connection member **70(c)** (shown in FIG. **29**) before attaching the first handle half **14(d)(i)** and second handle half **14(d)(ii)** to the handle connection member **70(c)**; (b) while the attachment ring **160** surrounds the lower portion of the handle connection member connecting the two handle halves **14(d)(i)**, **14(d)(ii)** to one another and to the handle connection member **70(c)**; (c) expanding the attachment ring **160** so that the inner diameter of the attachment ring **160** is increased enough to fit over the portion of the shaft below the circumferential recess **168**, (d) moving the expanded attachment ring **160** up into vertical alignment with the circumferential recess **168** of the shaft **35**; and (e) allowing the attachment ring **160** to return to its unexpanded state, thereby reducing the inner diameter of the attachment ring **160** such that the inner portion **162** of the attachment ring **168** is received in the circumferential recess **168**. Option “B” includes the following steps: (a) expanding the attachment ring **160** so that the slit **166** of the attachment ring **160** is wide enough to fit over the narrowest portion of the shaft **35**; (b) further expanding the attachment ring **160** so that the inner diameter of the attachment ring **160** is increased enough to fit over the lower portion of the shaft **35** just above the circumferential recess **168**, (c) moving the expanded attachment ring **160** down into vertical alignment with the circumferential recess **168** of the shaft **35**; and (d) allowing the attachment ring **160** to return to its unexpanded state, thereby reducing the inner diameter of the attachment ring **160** such that the inner portion **162** of the attachment ring **168** is received in the circumferential recess **168** of the shaft **35**. Option “C” includes the following steps: (a) aligning the attachment ring **160** with the circumferential recess **168** in the shaft **35**; (b) expanding the attachment ring **160** so that the slit **166** of the attachment ring **160** is wide enough to permit the attachment ring **160** to slide over shaft **35** at the circumferential recess **168**; (c) inserting the attachment ring **160** in the circumferential recess **168**, and (d) allowing the attachment ring **160** to return to its unexpanded state, thereby reducing the inner diameter of the attachment ring **160** such that the inner portion **162** of the attachment ring **168** is received in the circumferential recess **168**. Option “D” includes the following steps: (a) before attaching the steering handle **14(d)** to the steering controller **44(d)**, sliding the attachment ring **160** up around the base **54**; (b) expanding the attachment ring **160** so that the inner diameter of the attachment ring **160** is increased enough to fit over the base **54** and lower portion of the shaft **35**, (c) moving the expanded attachment ring **160** up into alignment with the circumferential recess

168 of the shaft **35**; and (e) allowing the attachment ring **160** to return to its unexpanded state, thereby reducing the inner diameter of the attachment ring **160** such that the inner portion **162** of the attachment ring **168** is received in the circumferential recess **168**.

(106) Once the attachment ring **160** has been installed on the shaft **35**, the upper opening in the flexible protective cover **33** can be widened by stretching/deforming the upper portion of the protective cover **33** so that it can be placed over, around, and in engagement with the outer portion **164** of the attachment ring **160**.

(107) In the embodiment depicted in FIGS. **29-31**, the upper inside portion of the protective cover **33** has an interior recess **170** (shown in FIGS. **29** and **31**) formed therein for receiving the outer portion **164** of the attachment ring **160**. When the upper portion of the protective cover **33** surrounds the attachment ring **160** and the outer portion **164** of the attachment ring **160** is received in interior recess **170** of the protective cover **33**, the portion of the protective cover **33** that surrounds the attachment ring **160** remains stretched/flexed/tensioned and exerts a radially inward compressive force on the attachment ring **160**. The inward compressive force imparted on the attachment ring **160** by the protective cover **33** can (1) eliminate or minimize the size of the slit **166** in the attachment ring **160**, (2) hold the inner portion **162** of the attachment ring **160** in the circumferential recess **168** of the shaft **35**, (3) hold the outer portion **164** of the attachment ring **160** in the interior recess **170** of the protective cover **33**, (4) inhibit rotation of the attachment ring **160** relative to the protective cover **33**, and (5) prevent the upper end of the protective cover **33** from sliding up or down relative to the shaft **35** of the steering handle **14(d)**.

(108) As shown in FIGS. **29-31**, the protective cover **33** can also include a lower circumscribing cover recess **37** for connecting the bottom of the protective cover **33** to the panel support structure **22** at the opening **23** in the panel support structure **22**. The flexibility of the protective cover **33** allows it to be attached to the panel support structure by simply (a) deforming the bottom of the protective cover **33** to have a smaller diameter than the opening **23**, (b) aligning the lower circumscribing cover recess **37** with the panel support structure **22**, (c) allowing the bottom of the protective cover **33** to expand to its normal shape with edges of the panel support structure **22** being received in the cover recess **37**, thereby attaching the bottom of the protective cover **33** to the panel support structure **22** in a manner that prevents dirt and debris from passing through the opening **23** and into contact with the steering controller **44(d)**.

Claims

1. A vehicle steering assembly for controlling movement of a vehicle having independently rotatable left and right ground-engaging traction elements, the vehicle comprising a control panel within reach of an operator of the vehicle, the control panel comprising a panel support structure for supporting one or more control or display mechanisms of the vehicle, the steering assembly comprising: a T-shaped steering handle extending above the panel support structure, wherein the steering handle comprises an upright shaft and a laterally extending crossmember rigidly coupled to the upright shaft; a steering controller positioned at least partly below the panel support structure, wherein the upright shaft comprises a base that actuates the steering controller; a flexible protective cover extending around and attached to the shaft, wherein the flexible protective cover is configured to prevent dust and debris from entering the controller at the base of the shaft, wherein the steering handle is shiftable relative to the panel support structure in forward and rearward directions to thereby cause corresponding forward and rearward rotation of both of the left and right traction elements, wherein the steering handle is rotatable relative to the panel support structure in clockwise and counterclockwise directions to thereby cause a change in the relative speeds and directions of rotation of the left and right traction elements, and an attachment ring for attaching the flexible protective cover to the shaft, wherein the flexible protective cover is stretched around the attachment ring to thereby exert a compressive force on the attachment ring.

2. The vehicle steering assembly of claim 1, wherein the shaft includes a circumferential recess for attaching the flexible protective cover to the shaft.
3. The vehicle steering assembly of claim 2, wherein an upper portion of the flexible protective cover is received in the circumferential recess to thereby attach the flexible protective cover to the shaft.
4. The vehicle steering assembly of claim 1, wherein, when the steering handle is rotated relative to the panel support structure, the flexible protective cover is not deformed.
5. The vehicle steering assembly of claim 1, wherein the attachment ring is positioned around the outside of the shaft and inside the flexible protective cover.
6. The vehicle steering assembly of claim 1, wherein the attachment ring is rotatably attached to the shaft.
7. The vehicle steering assembly of claim 1, wherein the shaft includes a circumferential recess for attaching the flexible protective cover to the shaft, wherein the attachment ring includes an inner portion received in the circumferential recess and an outer portion engaging the flexible protective cover.
8. The vehicle steering assembly of claim 7, wherein the flexible protective cover includes an upper internal recess, wherein the outer portion of the attachment ring is received in the upper internal recess of the flexible protective cover.
9. The vehicle steering assembly of claim 8, wherein the compressive force holds (i) the outer portion of the attachment ring in the upper internal recess of the flexible protective cover and (ii) the inner portion of the attachment ring in the circumferential recess of the shaft.
10. The vehicle steering assembly of claim 1, wherein the attachment ring comprises at least one of a polyamide and a fluoropolymer.
11. The vehicle steering assembly of claim 10, wherein the polyamide is nylon and the fluoropolymer is polytetrafluoroethylene (PTFE).
12. The vehicle steering assembly of claim 1, wherein the steering controller comprise a plurality of depressible buttons positioned generally below the steering handle, wherein the depressible buttons cooperatively control the speed and direction of rotation of the left and right traction elements, wherein the base of the shaft presents a lower surface having a curved topography, wherein the lower surface depresses the buttons when the steering handle is manipulated by the operator.
13. The vehicle steering assembly of claim 12, wherein the steering handle is biased toward a neutral position so that the steering handle automatically returns to the neutral position when not acted upon by the operator of the vehicle, wherein the depressible buttons include a front left button, a front right button, a rear left button, and a rear right button, wherein (i) moving the steering handle forward out of the neutral position, without twisting the steering handle, depresses the front left and front right buttons, (ii) moving the steering handle rearward out of the neutral position, without twisting the steering handle, depresses the rear left and rear right buttons, (iii) twisting the steering handle clockwise from the neutral position depresses the rear right and front left buttons, and (iv) twisting the steering handle counterclockwise from the neutral position depresses the front right and rear left buttons.
14. The vehicle steering assembly of claim 1, wherein the left and right traction elements each comprises tracks, wherein the vehicle is a compact utility loader.
15. A steering control assembly for controlling movement of a vehicle having left and right traction elements, the steering control assembly comprising: a steering handle comprising an upright shaft and a cross-member fixed to a top of the upright shaft, wherein the cross-member is configured for manual grasping by an operator of the vehicle; a steering controller actuated by the steering handle and configured to control movement of the vehicle based on the positioning of the steering handle by the operator; a flexible protective cover attached to the shaft and configured to prevent dust and debris from entering the steering controller; and an attachment ring for attaching the flexible

protective cover to the shaft, wherein the attachment ring comprises an inner portion and an outer portion, wherein the shaft defines a circumferential recess receiving the inner portion of the attachment ring, wherein the flexible protective cover defines an interior recess receiving the outer portion of the attachment ring, wherein the attachment ring acts as a bushing between the shaft and the flexible protective cover so that the shaft can be rotated without causing corresponding rotation of the flexible protective cover, wherein the steering handle is biased toward a neutral position so that the steering handle automatically returns to the neutral position when not acted upon by the operator of the vehicle, wherein the steering control assembly is configured such that (i) tilting the steering handle forward from the neutral position toward a forward position causes forward rotation of both the left and right traction elements, (ii) tilting the steering handle rearward from the neutral position toward a rearward position causes rearward rotation of both the left and right traction elements, (iii) twisting of the steering handle clockwise from the neutral position toward a clockwise twisted position causes forward rotation of the left traction element and rearward rotation of the right traction element, and (iv) twisting of the steering handle counterclockwise from the neutral position toward a counterclockwise twisted position causes forward rotation of the right traction element and rearward rotation of the left traction element.

16. The steering control assembly of claim 15, wherein the flexible protective cover is stretched over the attachment ring and exerts a compressive force on the attachment ring to thereby hold (i) the inner portion of the attachment ring in the circumferential recess of the shaft and (ii) the outer portion of the attachment ring in the interior recess of the of the flexible protective cover.

17. A vehicle steering assembly for controlling movement of a vehicle having independently rotatable left and right ground-engaging traction elements, the vehicle comprising a control panel within reach of an operator of the vehicle, the control panel comprising a panel support structure for supporting one or more control or display mechanisms of the vehicle, the steering assembly comprising: a T-shaped steering handle extending above the panel support structure, wherein the steering handle comprises an upright shaft and a laterally extending crossmember rigidly coupled to the upright shaft; a steering controller positioned at least partly below the panel support structure, wherein the upright shaft comprises a base that actuates the steering controller; a flexible protective cover extending around and attached to the shaft, wherein the flexible protective cover is configured to prevent dust and debris from entering the controller at the base of the shaft, wherein the steering handle is shiftable relative to the panel support structure in forward and rearward directions to thereby cause corresponding forward and rearward rotation of both of the left and right traction elements, wherein the steering handle is rotatable relative to the panel support structure in clockwise and counterclockwise directions to thereby cause a change in the relative speeds and directions of rotation of the left and right traction elements, and an attachment ring for attaching the flexible protective cover to the shaft, wherein the attachment ring comprises at least one of a polyamide and a fluoropolymer.

18. The vehicle steering assembly of claim 17, wherein the polyamide is nylon and the fluoropolymer is polytetrafluoroethylene (PTFE).

19. A vehicle steering assembly for controlling movement of a vehicle having independently rotatable left and right ground-engaging traction elements, the vehicle comprising a control panel within reach of an operator of the vehicle, the control panel comprising a panel support structure for supporting one or more control or display mechanisms of the vehicle, the steering assembly comprising: a T-shaped steering handle extending above the panel support structure, wherein the steering handle comprises an upright shaft and a laterally extending crossmember rigidly coupled to the upright shaft; a steering controller positioned at least partly below the panel support structure, wherein the upright shaft comprises a base that actuates the steering controller; a flexible protective cover extending around and attached to the shaft, wherein the flexible protective cover is configured to prevent dust and debris from entering the controller at the base of the shaft, wherein the steering handle is shiftable relative to the panel support structure in forward and rearward

directions to thereby cause corresponding forward and rearward rotation of both of the left and right traction elements, wherein the steering handle is rotatable relative to the panel support structure in clockwise and counterclockwise directions to thereby cause a change in the relative speeds and directions of rotation of the left and right traction elements, and an attachment ring for attaching the flexible protective cover to the shaft, wherein the shaft includes a circumferential recess for attaching the flexible protective cover to the shaft, wherein the attachment ring includes an inner portion received in the circumferential recess and an outer portion engaging the flexible protective cover, wherein the flexible protective cover includes an upper internal recess, wherein the outer portion of the attachment ring is received in the upper internal recess of the flexible protective cover.
